

Due date: Wednesday 8:50am, Oct. 7, 2026

Any function, library and API in OpenCV is not allowed to use.

1. Zooming and Shrinking (c/c++) (40%)

- (1) To optimize storage space for archival purposes, reduce the resolution of the original image *lena512_0.raw* with a ratio 1:0.7(358x358) using nearest neighbor interpolation and bilinear interpolation. Check and discuss your result with or without applying low-pass filter (use XnView) before shrinking. (figure, 5%; discussion, 5%)
- (2) In preparation for high-resolution exhibits or analysis, upscale the original *lena512_0.raw* image to 1024x1024 using nearest neighbor interpolation and bilinear interpolation. Display the upscaled image. (figure, 10%)
- (3) In practical applications, such as high-speed cameras and surveillance systems, image resolution may be limited by hardware constraints. Multi-frame super-resolution (MFSR) can overcome this limitation by combining multiple low-resolution images to reconstruct a higher-resolution image.

Use four low-resolution images, *lena512_0.raw*, *lena512_1.raw*, *lena512_2.raw*, and *lena512_3.raw*, to perform MFSR shift-and-add with an enlargement factor of 2.

Reconstruct a high-resolution image of size 1024×1024 pixels. (figure, 10%, discussion, 5%)

Equation 1: Sampling Region

$$\begin{cases} LR_0: (\Delta y_0, \Delta x_0) = (0,0) \\ LR_1: (\Delta y_1, \Delta x_1) = (0,1) \\ LR_2: (\Delta y_2, \Delta x_2) = (1,0) \\ LR_3: (\Delta y_3, \Delta x_3) = (1,1) \end{cases}$$

Equation 2: Spatial Offsets

$$\Omega_k(x, y) = \{ (X, Y) \mid X \in \{2x + \Delta x_k, 2x + \Delta x_k + 1\}, Y \in \{2y + \Delta y_k, 2y + \Delta y_k + 1\} \}$$

Equation 3: Reconstruct

$$HR(X, Y) = \frac{1}{|S(X, Y)|} \sum_{(k, x, y) \in S(X, Y)} LR_k(x, y)$$

$$S(X, Y) = \left\{ (k, x, y) \mid x = \left\lfloor \frac{X - \Delta x_k}{2} \right\rfloor, y = \left\lfloor \frac{Y - \Delta y_k}{2} \right\rfloor, 0 \leq x < W_{LR}, 0 \leq y < H_{LR} \right\}$$

- (4) Calculate the PSNR of the three enlarged images obtained in (2) and (3), using *lena1024.raw* as the reference image. Present the results in a comparison table and discuss the differences in reconstruction quality among nearest neighbor interpolation, bilinear interpolation, and multi-frame super-resolution. (Results, 3%; Discussion, 2%)

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$$

$$PSNR = 10 \times \log_{10} \left(\frac{MAX_I^2}{MSE} \right) = 20 \times \log_{10} \left(\frac{MAX_I}{\sqrt{MSE}} \right)$$

2. False Contouring and Dithering (C/C++) (30%)

Reduce the grayscale resolution of the image *duck900x660.raw* from 8 bits to 4 bits and 1 bit through quantization. Next, implement the **Bayer-Ordered dithering algorithm** (study yourself) using a **4×4 Bayer threshold matrix** to spread the quantization error across neighboring pixels, thereby reducing the visibility of these false contours. Finally, compare the quantized image (without dithering) and the dithered image with the origin image. In your discussion, explain how dithering reduces false contouring and improves perceived image quality. (figure, 20%, discussion, 10%)

$$B_4 = \begin{bmatrix} 0 & 8 & 2 & 10 \\ 12 & 4 & 14 & 6 \\ 3 & 11 & 1 & 9 \\ 15 & 7 & 13 & 5 \end{bmatrix}$$

4×4 Bayer threshold matrix



duck900x660.raw

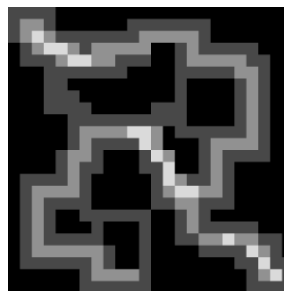
3. Shortest Path Detection Using Distance Metrics (C/C++) (30%)

In this assignment, you will use D4 and Dm distance metrics to find the shortest path of the ball in the image file *map24x24.raw*, then replace the path with the color white. The starting point of the shortest path should be from start (0,0) to end (23,23) on gray-value of the road {72}.

Compare the results and explain the differences between D4 and Dm in terms of pixel adjacency.

(1) D4 (figure ,10%, discuss, 5%)

(2) Dm (figure ,10%, discuss, 5%)



map24x24.raw